

System Requirements for Study Assistant Application

Functional Requirements:

1. The application shall allow users to upload study materials in PDF, DOCX, or text format.
 2. The application shall analyze uploaded study materials and generate:
 - Summaries of the content.
 - Key concepts and important topics.
 - Practice quiz questions.
 3. The application shall provide quiz functionality including:
 - Multiple choice questions.
 - True/False questions.
 - Short answer questions.
 4. The system shall track user quiz performance and identify weak subject areas.
 5. Users shall be able to review previously uploaded study materials and generated quizzes.
 6. The application shall provide personalized study recommendations based on quiz performance and topic weaknesses.
 7. The application shall allow users to toggle between light mode and dark mode.
 8. Users shall be able to search previously uploaded study materials by title or keyword.
 9. The application shall provide an AI-powered chatbot to answer questions related to uploaded study materials.
 10. The system shall automatically save user progress and quiz history.
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Performance Requirements:

1. The application shall process and summarize uploaded study materials within 20 seconds for files under 25 MB.
 2. The system shall generate quizzes within 10 seconds after analysis is complete.
 3. AI-generated responses shall be delivered within 10 seconds under normal network conditions.
 4. The application shall support up to 500 concurrent users.
 5. Background data synchronization shall occur without disrupting active user sessions.
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Platform Requirements:

- Laptop and desktop devices only.
- Modern web browsers including:
 - Google Chrome
 - Mozilla Firefox
 - Microsoft Edge
 - Safari

The application shall support screen resolutions of 1280x720 and above.

Data Requirements:

1. The application shall use third-party AI APIs and NLP libraries for content analysis and quiz generation.
 2. User preferences and settings shall be stored securely in a database or local device storage.
 3. Uploaded study materials shall be securely stored and associated with user accounts.
 4. Quiz scores, study history, and topic performance analytics shall be maintained for progress tracking.
 5. Frequently accessed study materials shall be cached temporarily to improve performance.
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Security Requirements:

1. All API communications shall use HTTPS encryption.
 2. The application shall not store sensitive user information without user consent.
 3. Uploaded study materials shall only be accessible to the user who uploaded them.
 4. API keys and authentication credentials shall be securely stored on the backend server.
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Usability Requirements:

1. The application shall provide a simple and user-friendly interface for students.
 2. Navigation between uploads, summaries, quizzes, and analytics shall be intuitive.
 3. Error messages shall clearly explain issues and possible resolutions.
 4. Users shall be able to easily access recent uploads and quiz history.
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Reliability Requirements:

1. The system shall maintain 99.9% uptime over a 30-day period.
2. The application shall retry failed API requests up to three times before displaying an error message.
3. The system shall automatically save user progress during active sessions.
4. Cached study materials shall remain accessible during temporary internet interruptions.
5. The application shall gracefully recover from unexpected crashes without data loss.